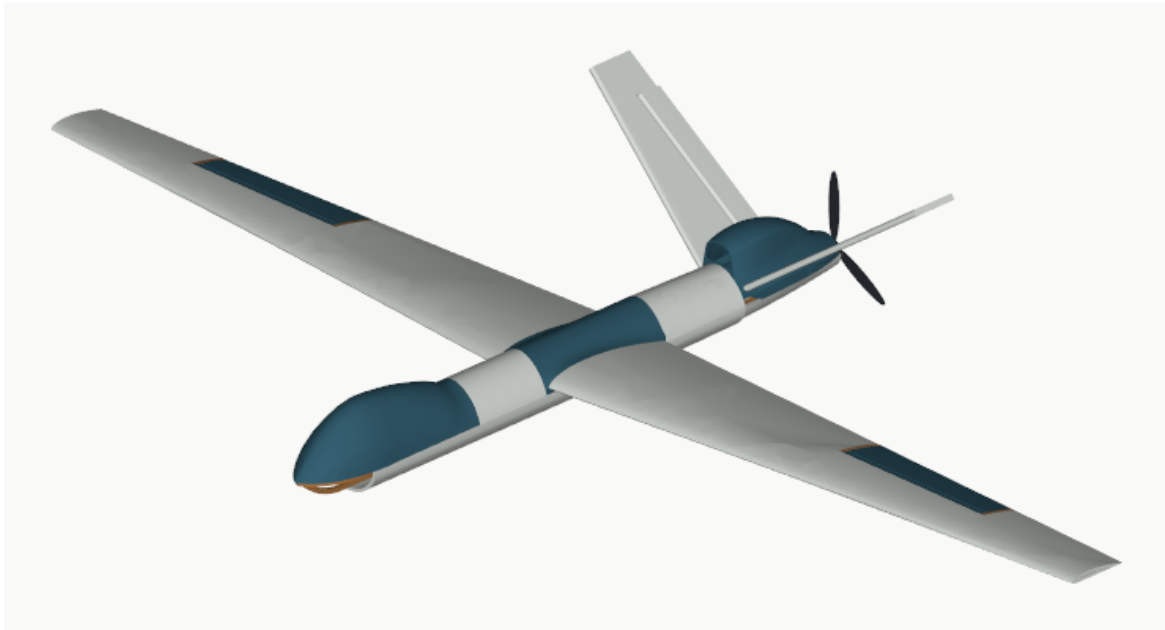


Build it in this order



A practical assembly guide for the original Michael Rehtin / Onshape airframe, adapted to owned 6 mm plywood, PLA+, 5 mm foamboard and two whole carbon rods. CAD illustrations show intended placement; they are not photographs of a built aircraft.

Hero: motor and 9-inch three-blade propeller are dimensional illustrations. Landing gear is not shown because the purchased set and attachment are not yet selected.

Start here: performance and preparation

Cutting, printing and part identification

Assemble the frame and wings

Foam, controls and propulsion

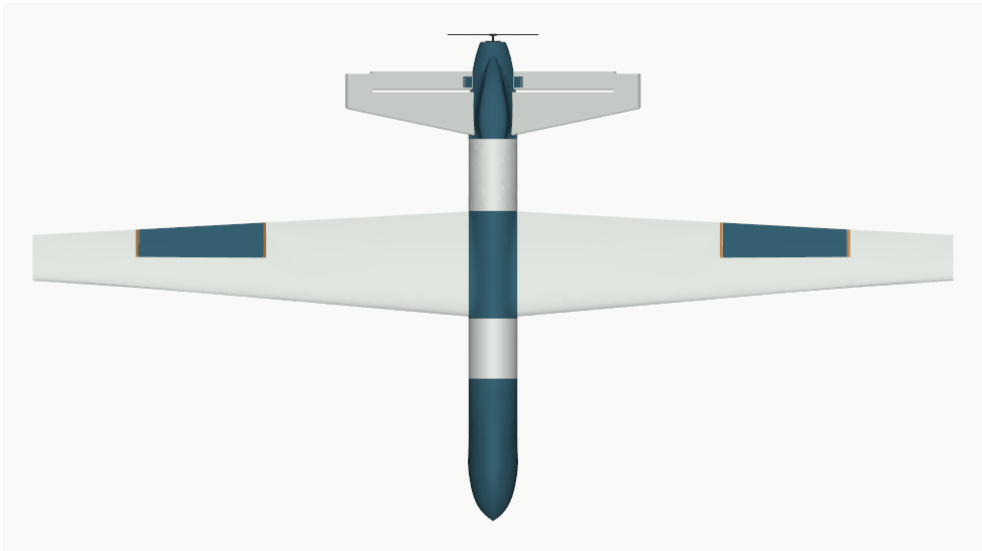
Place and wire the electronics

Configure, test and prepare for flight

File index and sources

Build and bench guide. Check the issued part and panel manifests before fabrication. Physical fit, bond strength, balance, propulsion and pilot acceptance remain required.

Know the aircraft first



Measure / estimate	Value	Basis
Span / wing area	2.00 m / 0.304 m2	Original-source lifting geometry; exposed panels.
Flying mass	2.51 kg estimate	Assumed densities, sliced PLA+, 512 g battery, future gear allowance; range 2.11–3.07 kg. Weigh built aircraft.
Wing loading	83 g/dm2 estimate	Mass estimate divided by 0.304 m2 exposed area.
Balance starting check	Y 74 mm (20% chord)	Geometric starting target, not a validated stability limit. At battery Y=-260 estimate is Y60; needed battery Y=-192 clips the front former. Battery Y=-200 gives about Y72.5 (19% chord); weigh and test actual access.
Stall speed	38–49 km/h scenario	At 2.51 kg, assumed CLmax 0.8–1.2 and air density 1.1–1.225 kg/m3. Not measured.
Launch / runway	Not established	Illustrative 1.3x stall is 49–63 km/h. Ideal 39–156 m at assumed net 4–1 m/s2; thrust, rolling drag, rotation and gear unmeasured.
Flight time	8–25 min scenario	80% of 5.2 Ah at measured average 30–10 A. Example 15 A gives 16.6 min. No current test yet.

Calculate from the finished R5 geometry, then weigh and balance the aircraft in its actual flying configuration. Predictions do not prove glue-joint strength, propulsion margin, ground clearance or flight stability.

Prepare the bench

Have ready	Use
Flat board, ruler, square, calipers	Keep the spine straight, match left/right wings, measure stock and fits.
Sharp knife, spare blades, straightedge, sanding block	Cut foam cleanly; bevel and relieve the inner face without damaging the outside paper.
Hand drill, small bits, smooth trial pin	Finish pilot holes and verify motor screws; drill only after the real part agrees with the template.
Clamps, pins, tape, disposable spacers	Hold dry assemblies and cured bonds without twisting the frame.
Foam-compatible adhesive, wood glue, tape, zip ties	Use a tested adhesive for each material. Tape/ties provide simple, inspectable equipment restraint.
Soldering tools, heat-shrink, multimeter, current-limited supply	Inspect joints and test continuity/polarity before connecting the flight battery.

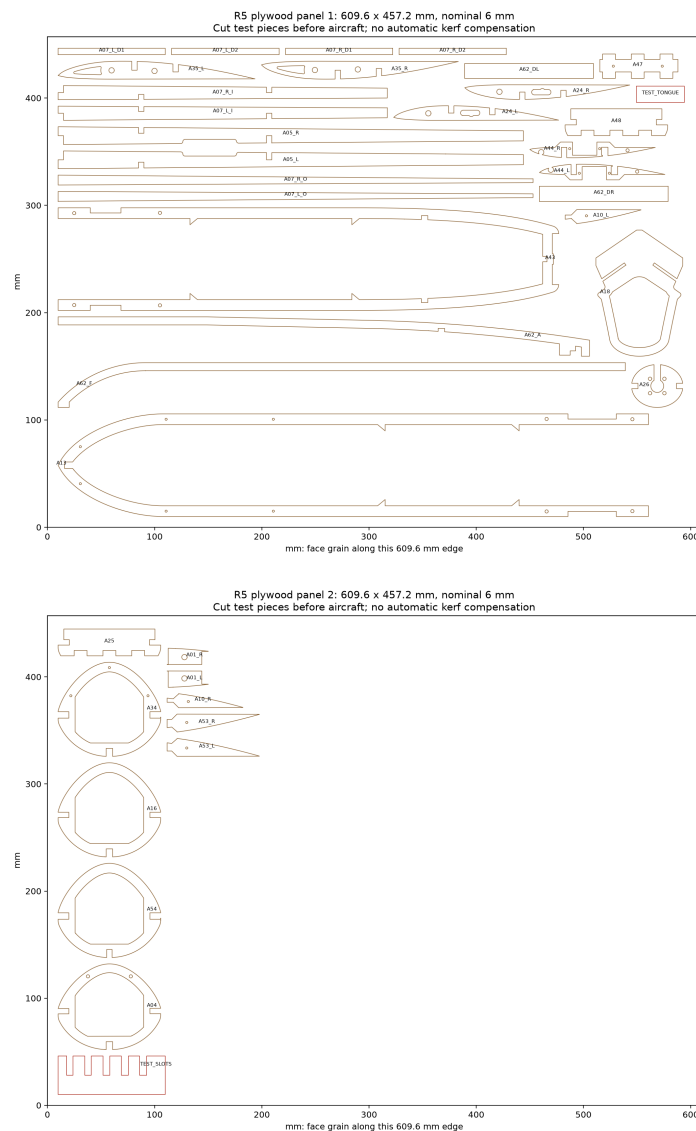
Test the materials before full parts

Measure plywood and foam thickness at several places. Cut a kerf/slot coupon with the actual shop settings. Test a wood joint, a foam fold and each adhesive on offcuts, then inspect after full cure. Original video wood/superglue joints failed: do not treat a neat appearance as structural proof.

Ordinary CA must not contact exposed XPS foam unless its manufacturer explicitly approves that use. Avoid solvent glues that attack the core. Hot glue can locate light equipment, but it is not the only restraint for the flight battery or a loaded joint.

Label every part before it leaves the sheet. Right and left refer to the pilot looking forward. In CAD: X is across the wings, Y increases toward the tail, and Z is upward. Keep a small bag for each stage.

Cut the plywood



01 Use the actual panel drawings

Use only the R5 workshop panel DXFs. Stock is 609.6 x 457.2 mm, with face grain along the 609.6 mm edge. Cut in millimetres at 100%; never resize a part to make it fit a sheet.

02 Qualify the cut first

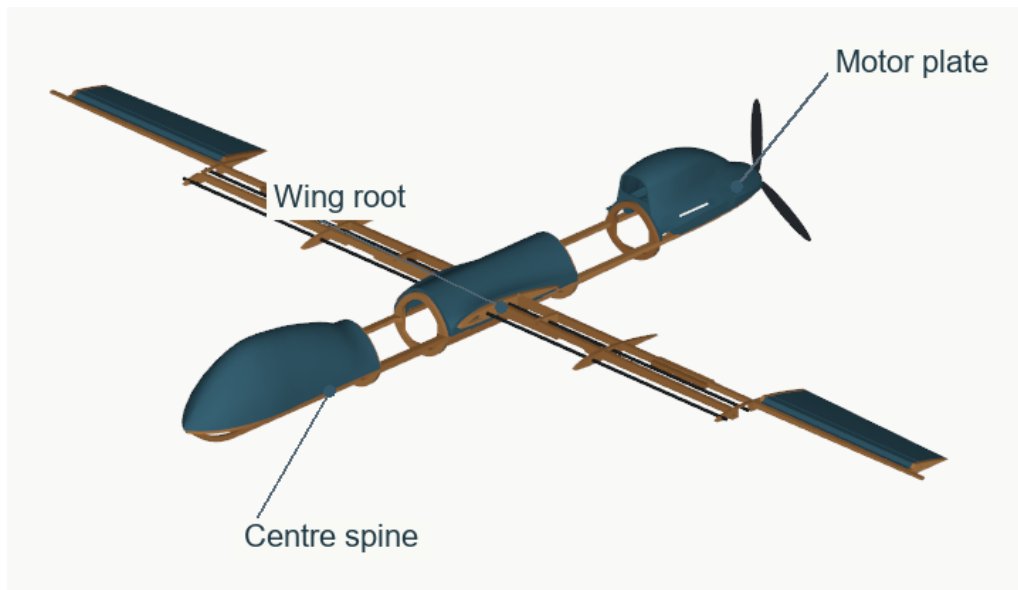
Measure the coupon and actual slot fit before cutting the full sheets. The shop must choose kerf compensation from that result. Deburr lightly; do not round away load-bearing tabs or change hole centres.

03 Sort and mark

There are 37 wooden aircraft pieces plus 2 test coupons on 2 panels. Coupons are not installed. Keep every ID and side mark. Long pieces use only the specified paired laps. Butt ends locate the parts; the two laps carry the joint. Do not omit a lap because the frame appears rigid.

Print and inspect

The assembly list contains 14 new printed pieces and 1 reused nose piece. Print any fit coupons separately; they are test tools, not installed aircraft parts. Optional original gear prints stay on hold.



04 Reuse the nose

N1 / A59 is already printed. Reuse that exact part. Do not reprint or scale it. Confirm clearance before making the local pitot passage.

05 Confirm the printer envelope

The printer model is not identified. The only confirmed observation is that the nose fit with about 20 mm side clearance. Open each arranged job in the actual printer slicer and check bed, height, supports and travel limits. Original ailerons/rails are long; upright height can be about 274 mm.

06 Use PLA+ and trial fits

Start with the supplied oriented parts, then choose wall, infill and support settings for the actual machine. Print the fit coupon first. Fit both whole 5 mm rods in the nominal 5.2 mm printed bores. Finish locally if required; never scale a root or control surface.

07 Clean and inspect

Remove support from sockets and hinge pilots. Reject cracked, warped or poorly bonded prints. Weigh finished parts and record them. Optional original gear pieces remain on hold until the bought gear and attachment method are settled.

A printed infill percentage is not a strength rating. Layer direction, walls, material, temperature and joint loading all affect the finished part.

Plywood part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



A04 | 1 pc

Rear body former



A13 | 1 pc

Front horizontal fuselage frame



A16 | 1 pc

Body former at Y233.65mm



A18 | 1 pc

Tail-carrier bulkhead



A25 | 1 pc

Forward main-gear crossmember



A26 | 1 pc

Motor firewall



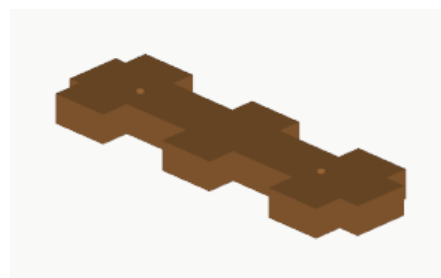
A34 | 1 pc

Front body former at Y-130mm



A43 | 1 pc

Aft horizontal fuselage frame



A47 | 1 pc

Main-gear crossplate



A48 | 1 pc

Aft main-gear crossmember

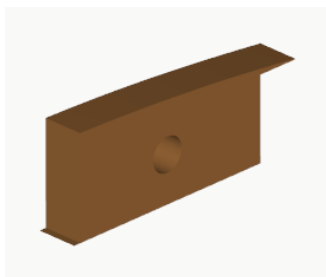
Plywood part identification

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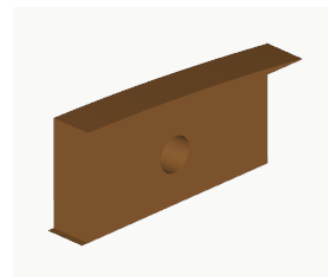
A54 | 1 pc

Body former at Y-5mm



A01_R | 1 pc

Rear-rod support rib at span130mm - right



A01_L | 1 pc

Rear-rod support rib at span130mm - left



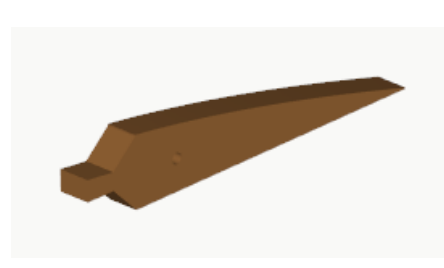
A05_R | 1 pc

Front wing spar - right



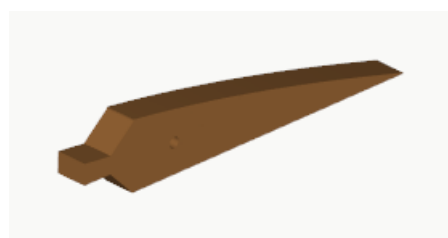
A05_L | 1 pc

Front wing spar - left



A10_R | 1 pc

Outer aileron endplate - right



A10_L | 1 pc

Outer aileron endplate - left



A24_R | 1 pc

Middle wing rib at span250mm - right



A24_L | 1 pc

Middle wing rib at span250mm - left



A35_R | 1 pc

Inner wing rib at span53mm - right

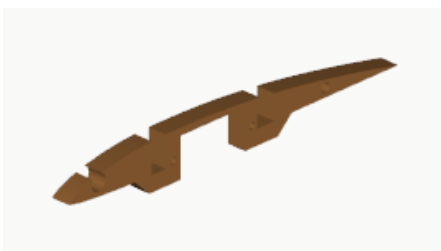
Plywood part identification

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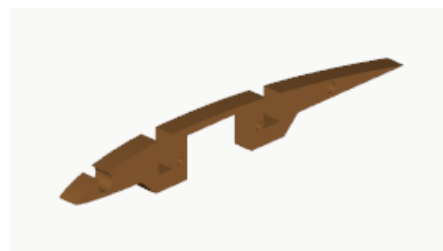
A35_L | 1 pc

Inner wing rib at span 53mm - left



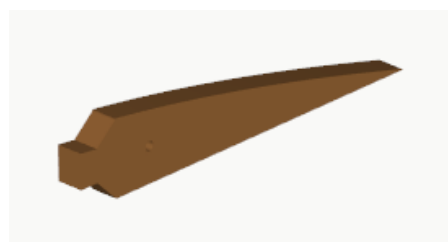
A44_R | 1 pc

Outer wing rib at span 490mm - right



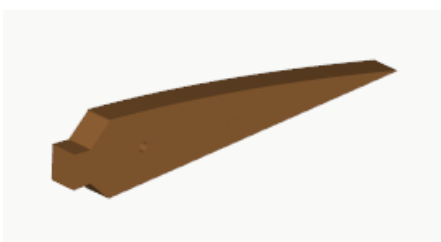
A44_L | 1 pc

Outer wing rib at span 490mm - left



A53_R | 1 pc

Inner aileron endplate - right



A53_L | 1 pc

Inner aileron endplate - left



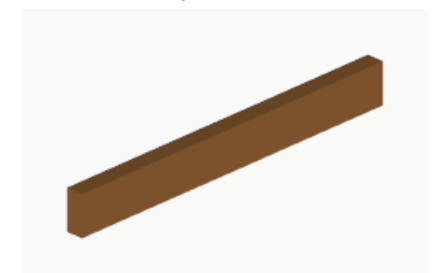
A62_F | 1 pc

Lower fuselage keel - front half



A62_A | 1 pc

Lower fuselage keel - aft half



A62_DL | 1 pc

Keel joint - left side lap



A62_DR | 1 pc

Keel joint - right side lap



A07_R_I | 1 pc

Rear wing spar - right inboard section

Plywood part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



A07_R_O | 1 pc

Rear wing spar - right outboard section



A07_R_D1 | 1 pc

Rear wing spar - right first face lap



A07_R_D2 | 1 pc

Rear wing spar - right second face lap



A07_L_I | 1 pc

Rear wing spar - left inboard section



A07_L_O | 1 pc

Rear wing spar - left outboard section



A07_L_D1 | 1 pc

Rear wing spar - left first face lap



A07_L_D2 | 1 pc

Rear wing spar - left second face lap

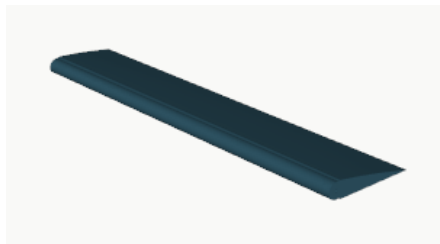
PLA+ part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



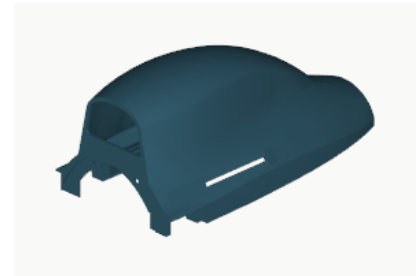
A06 | 1 pc

Centre fairing



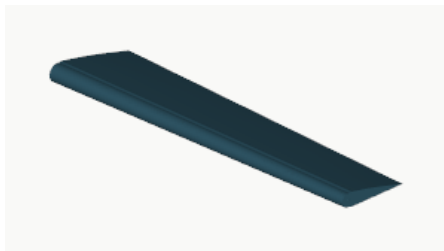
A11 | 1 pc

Left printed aileron



A12 | 1 pc

Tail carrier



A21 | 1 pc

Right printed aileron



A23 | 1 pc

Left fixed aileron hinge rail



A36 | 1 pc

Right fixed aileron hinge rail



A45 | 1 pc

Right wing root



A55 | 1 pc

Left wing root

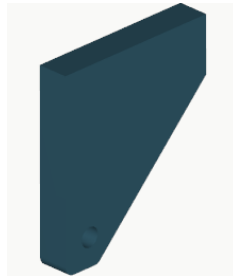
PLA+ part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



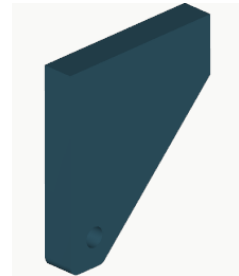
A59 | 1 pc

Owned N1 nose, reused



HWR | 1 pc

Right aileron horn



HWL | 1 pc

Left aileron horn



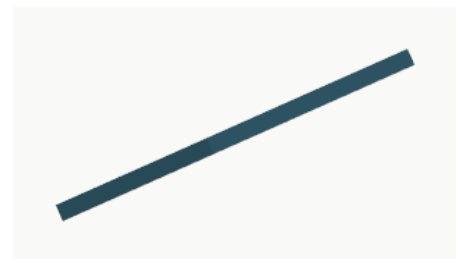
HTR | 1 pc

Right tail horn with base



HTL | 1 pc

Left tail horn with base



HTRB | 1 pc

Right tail horn backing plate

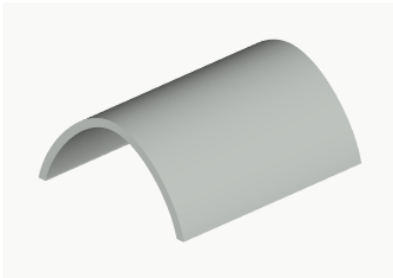


HTLB | 1 pc

Left tail horn backing plate

Foam part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



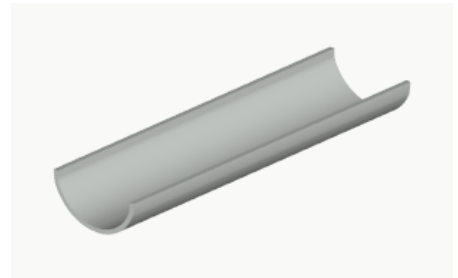
A08 | 1 pc

Forward upper body foam



A15 | 1 pc

Aft upper body foam



A17 | 1 pc

Middle lower body foam



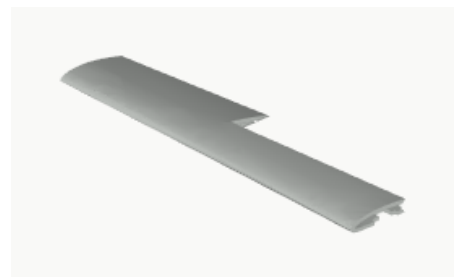
A20 | 1 pc

Tail lower body foam



A22 | 1 pc

Left fixed V-tail foam



A27 | 1 pc

Left outer wing foam



A38 | 1 pc

Right outer wing foam

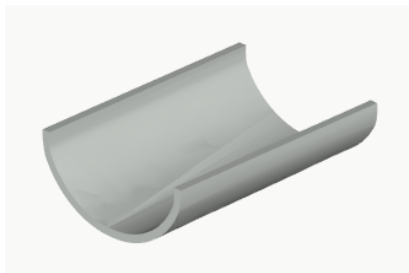
Foam part identification

Every item shown has quantity 1 unless its caption states otherwise. Source IDs identify placed CAD parts; foam cutting-pattern IDs may combine several surfaces into one wrap.



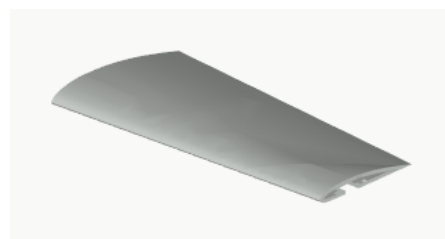
A41 | 1 pc

Right moving V-tail foam



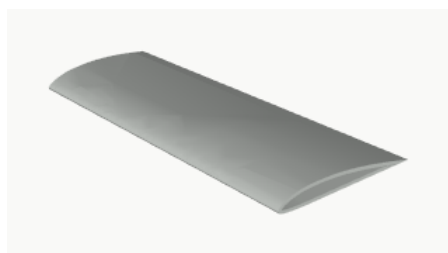
A42 | 1 pc

Aft lower body foam



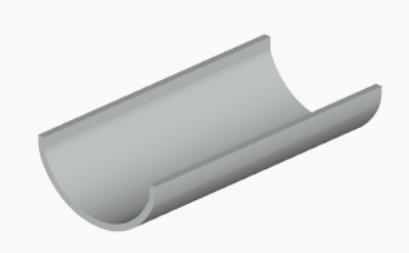
A49 | 1 pc

Right inner wing foam



A52 | 1 pc

Left inner wing foam



A56 | 1 pc

Forward lower body foam



A57 | 1 pc

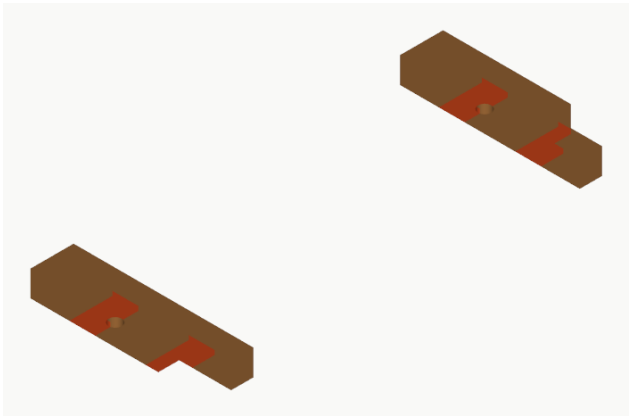
Right fixed V-tail foam



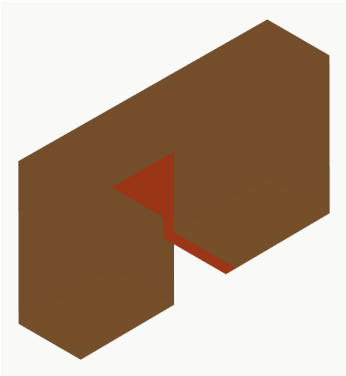
A60 | 1 pc

Left moving V-tail foam

Finish only the marked mating faces



A43 detail from actual CAD. Orange is local material to remove, not an added part. The two gear-bay mating patches lose 1 mm from the bottom face, leaving 5 mm. Surrounding stock remains 6 mm.

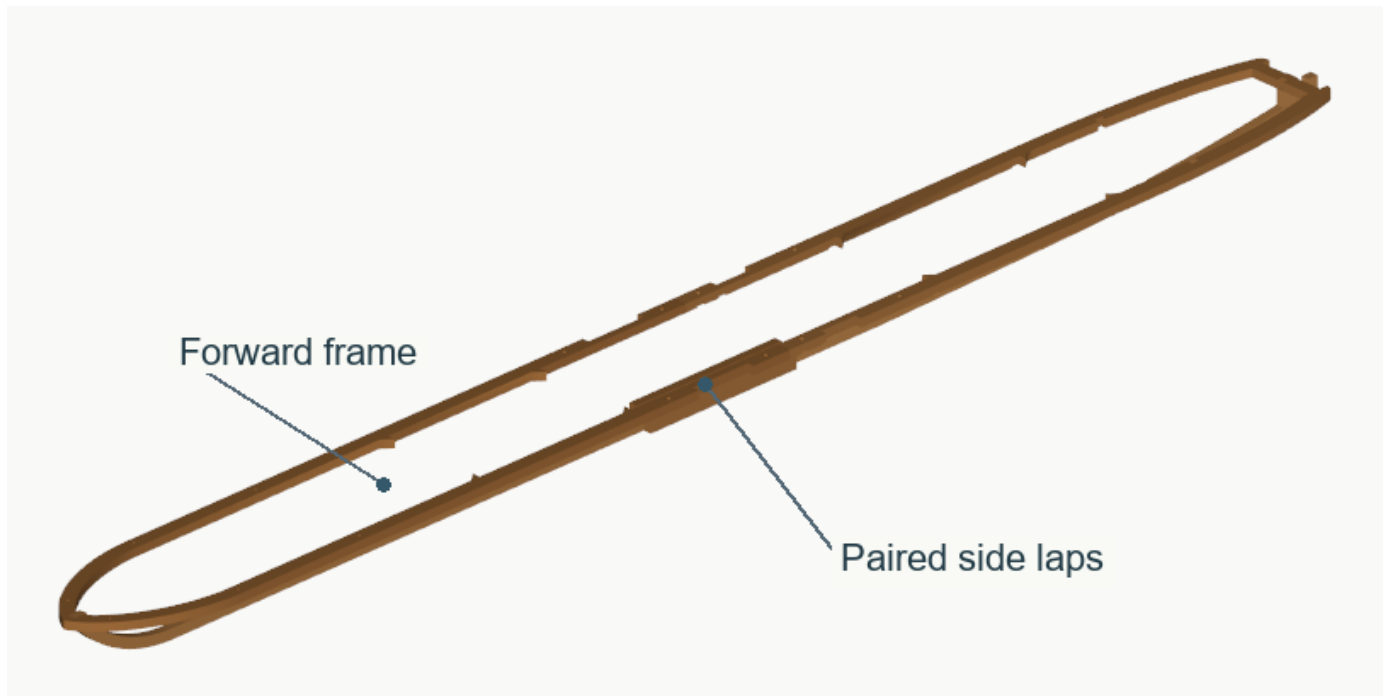


A05_R front-spar detail. Notch/edge relief follows the actual mating rib. A bounding rectangle is not a command to remove its whole area or depth. Left side is mirrored.

Where	What to finish
A43 at gear bay	Remove 1 mm from the bottom only at the two marked patches. Check 5 mm remaining with calipers.
A43 at A16/A04; A05_R/L; A07_R/L inner sections	Pare the marked mating faces to the removal-solid detail; repeatedly dry-fit the actual mating part. Preserve full-thickness bridges.
A62_F at A54	Finish the local centre-former notch shown in the removal detail. Keep both sides aligned.
A53_R/L and outer rear-spar ends	Only local edge finishing is intended; do not thin the entire end plate or spar.

Mark the intended contact, take small cuts, check with the mating part and stop as soon as it seats without force. Consult the supplied secondary-removal STEP details for local shapes. Never plane every 6 mm piece down to 5 mm.

Join the centre spine



08 Lay out the spine

Place A62_F and A62_A on a flat board, aligned along the body centreline. Their seam is at CAD Y = 100 mm. Match the source-shaped upper and lower edges before bonding.

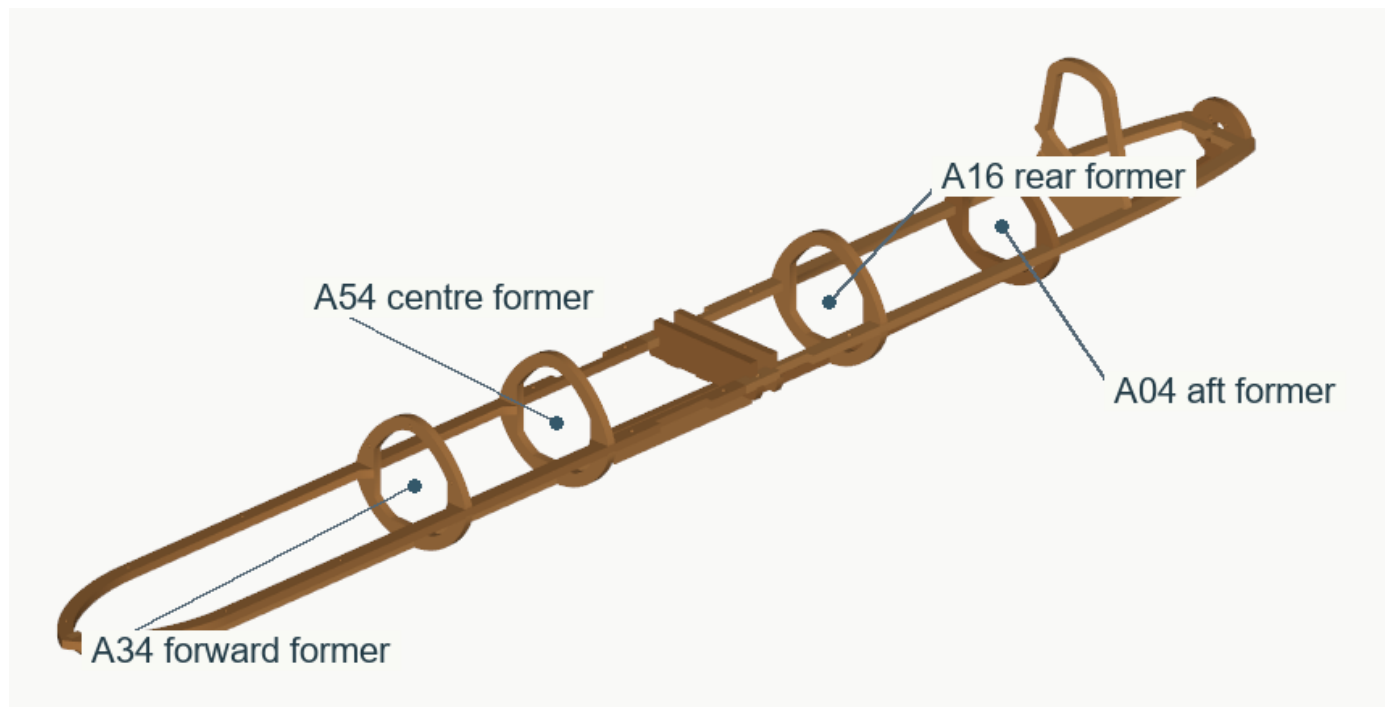
09 Fit both side laps

Fit A62_DL and A62_DR, one on each side. Each lap is 120 mm long and covers 60 mm of each end. Their lower edges sit at CAD Z = -41.714 mm. Dry-fit the surrounding parts first, then bond the complete contact faces and clamp flat until fully cured.

10 Add the horizontal frame

Dry-fit A13 forward and A43 aft. Keep the structure square and straight. Do not remove wood to hide a wrong orientation. A40 is incorporated into the unified spine footprint; it is not an extra piece to stack on it.

Build the fuselage frame



11 Seat the formers

Set the forward face of A34 as your ruler zero. A54 is 125 mm aft, A16 is 363.65 mm aft and A04 is 514.65 mm aft. Their slots and the placed assembly set position. Keep the horizontal frame upper face at Z0; seat tabs fully and check squareness.

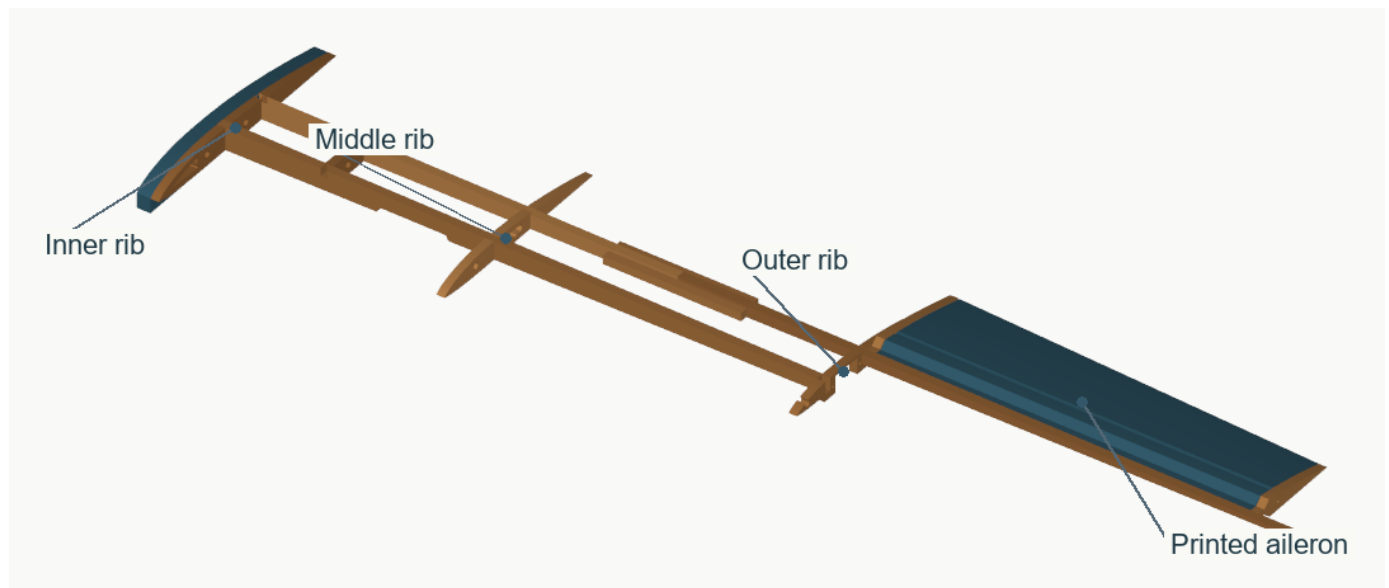
12 Fit the aft structure

Fit A18 tail bulkhead, A26 motor plate and A25/A47/A48 gear bay pieces. Trial-fit the printed nose, centre fairing and tail carrier around the full dry frame.

13 Bond without closing access

Bond only after the complete dry frame aligns. Preserve access to motor screws, battery restraint, flight controller, wiring connectors and tail linkages. Allow the specified adhesive cure before removing clamps.

Build matching wing frames



14 Sort right and left

Use the _R parts for the right wing and _L parts for the left. Assemble inner rib A35, middle rib A24 and outer rib A44 with front spar A05, rear spar A07 and rod support A01.

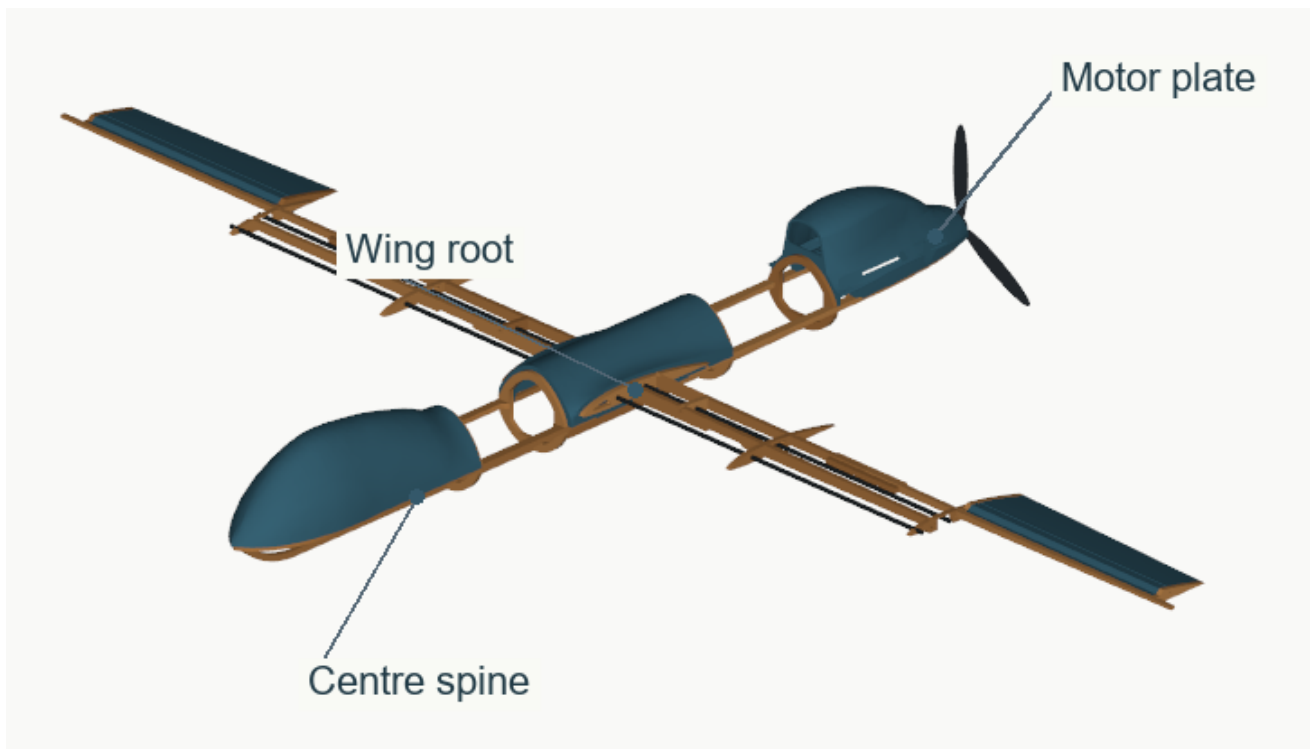
15 Finish each rear spar joint

Join A07*_I and A07*_O at $X = \pm 360$ mm. Fit both face laps D1 and D2: nominally 100 mm long, 50 mm each side of the seam. Bond full contact, keeping the spar straight.

16 Check symmetry before glue

Build on a flat support. Compare rib spacing, leading and trailing edges and incidence on both sides. Trial-fit the outer foam and aileron rail; finish only identified local interference before bonding.

Fit the whole carbon rods



17 Keep both rods whole

Use two solid 5 x 1000 mm rods centred across the aircraft, ends at X = -500 and +500 mm. One follows the front spanwise axis; the other follows the rear axis. Do not cut or butt them at the centre.

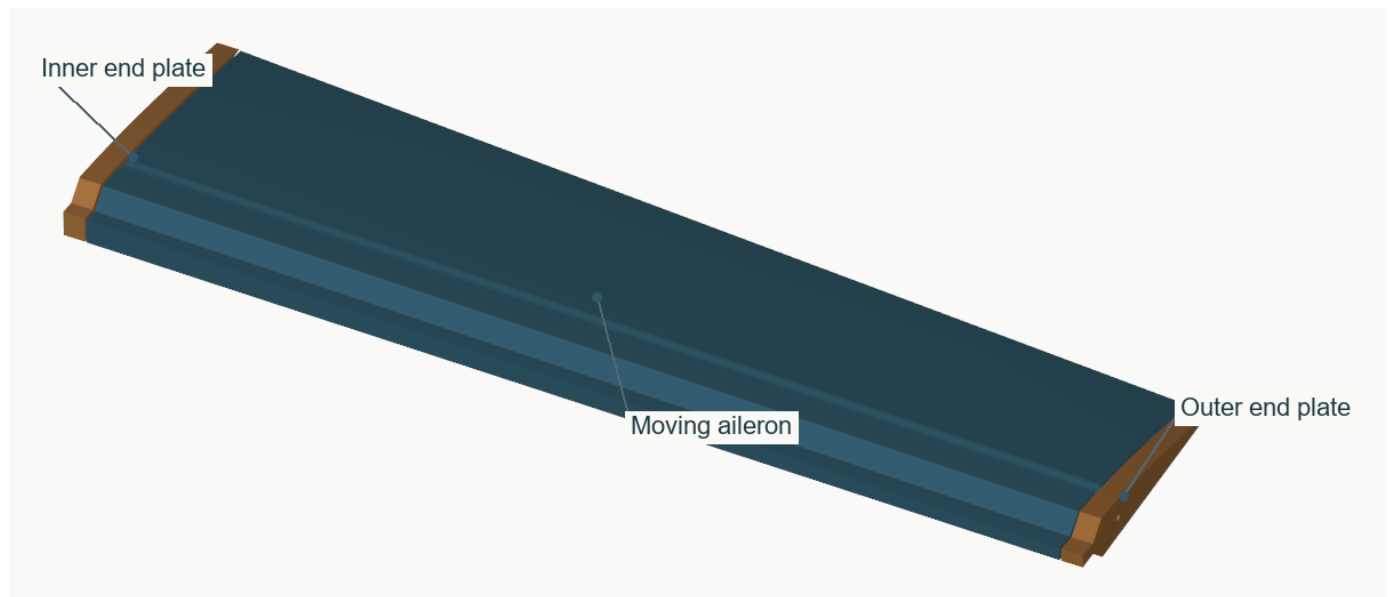
18 Slide through a complete dry assembly

Fit both printed roots and both wing frames together. Front axis: Y60.454, Z15.632 mm. Rear axis: Y100.500, Z15.070 mm. Check each wood opening and local inner-foam relief; preserve the exterior skin.

19 Prove fit and retention

Rod fit must be snug without splitting a print or forcing a rib. Check that neither wing nor rod can escape spanwise. Complete the specified root retention and structural trial before closing the skin; a resized hole alone does not establish strength.

Install the printed ailerons



20 Align the two end pivots

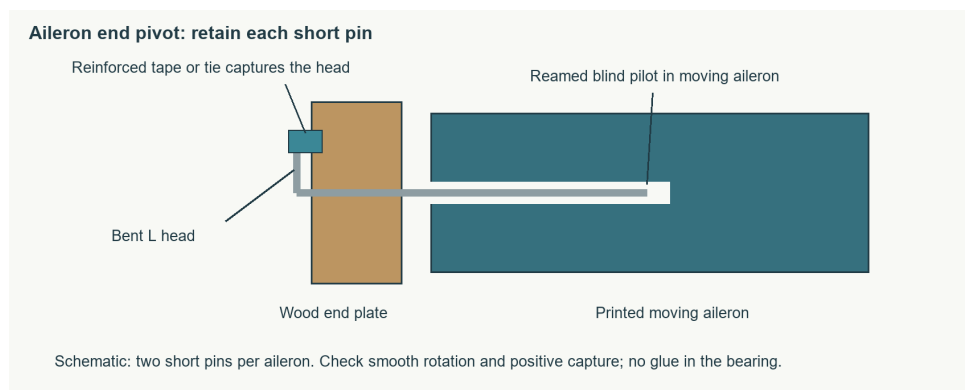
The R5 ailerons retain the original shape with corrected common-axis 2 mm pilot holes. Dry-fit A21/A36 with right end plates A53_R/A10_R; use A11/A23 and left plates on the other wing.

21 Fit short end pins

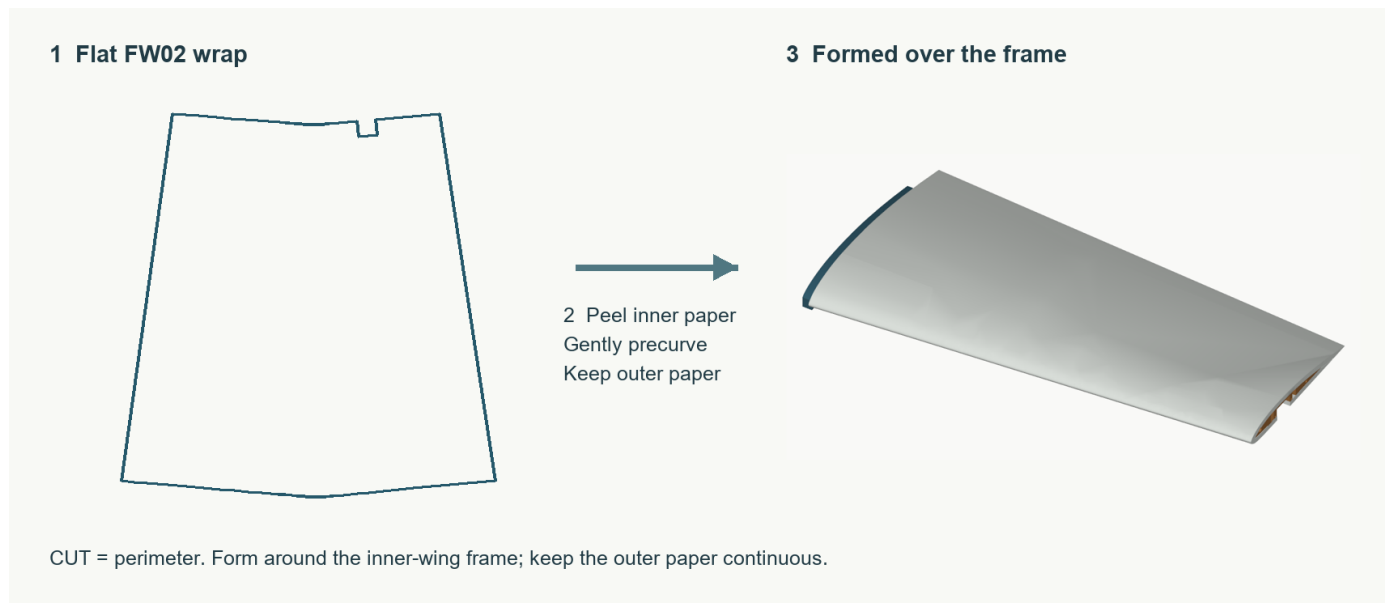
Measure the actual spoke/pin material. Carefully finish the pilots to a smooth rotating fit along the common hinge line. Each aileron uses two short end pins, not a full-length shaft. Keep the moving surface clear of the fixed rail.

22 Retain and test

Use short spoke-offcut pins with bent L heads. Trap each L head against its wooden end plate with reinforced tape or a tie so the pin cannot slide out; keep the head clear of the moving surface. Keep adhesive out of the bearing. Move the aileron through its full required travel; reject scraping, binding, excess play or a pin that can work free.



Form practical foam wraps



23 Mark the outside face

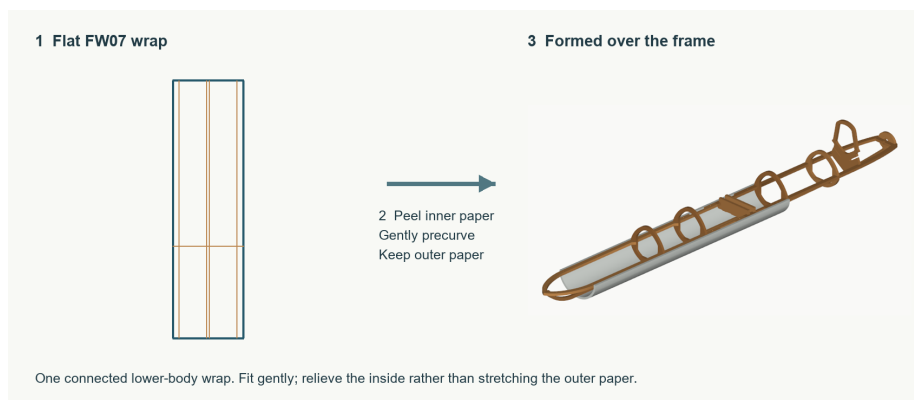
Use the simplified R5 wrap patterns. Keep the outer paper intact at the fold line; mark the ID, front edge, centre reference and inside face before cutting. Start with a representative foam/glue/fold coupon.

24 Dry-form over the frame

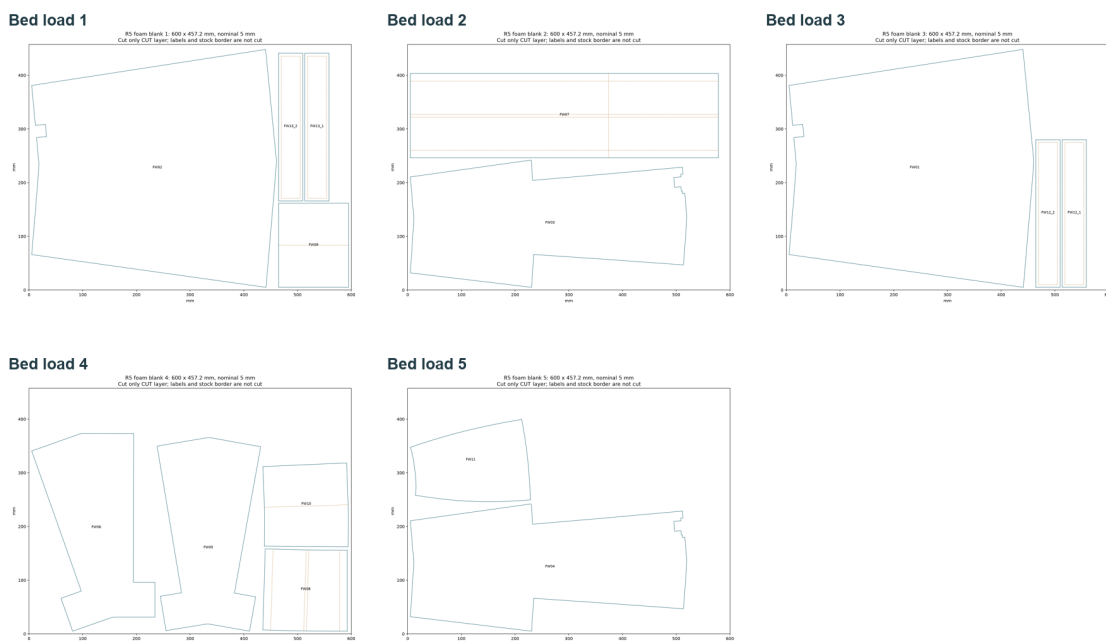
Curve progressively around the leading edge and frame. Remove or compress inner core only where the pattern calls for a fold, bevel or fitted relief. On the right outer wing FW03, pare inner foam at A10_R and A53_R wooden endplates by dry fit; preserve the complete outside paper and full span. Check the rods, ribs and hinge rail.

25 Close only after checks

Check the entire wing with its moving aileron, servo lead and linkage fitted. Bond progressively while holding the airfoil and symmetry. Tape the finished seam and protect exposed paper edges without bridging a moving hinge.



Foam pattern and sheet map



Wing wraps: FW01 left inner; FW02 right inner; FW03 right outer; FW04 left outer. At FW03, pare the inner foam locally against wooden A10_R and A53_R during dry fit. Keep the outside paper and full outer-wing span. **Fixed tails:** FW05 right; FW06 left. Each fixed tail is one whole skin including its root collar.

FW07: long lower body. One 157.4 x 573.65 mm wrap replaces the separate front/middle strips. Peel the inner paper and gently precurve; the 205 mm cross-line is a station reference. **FW08:** aft lower wrap. **FW09/FW10:** front/rear upper covers.

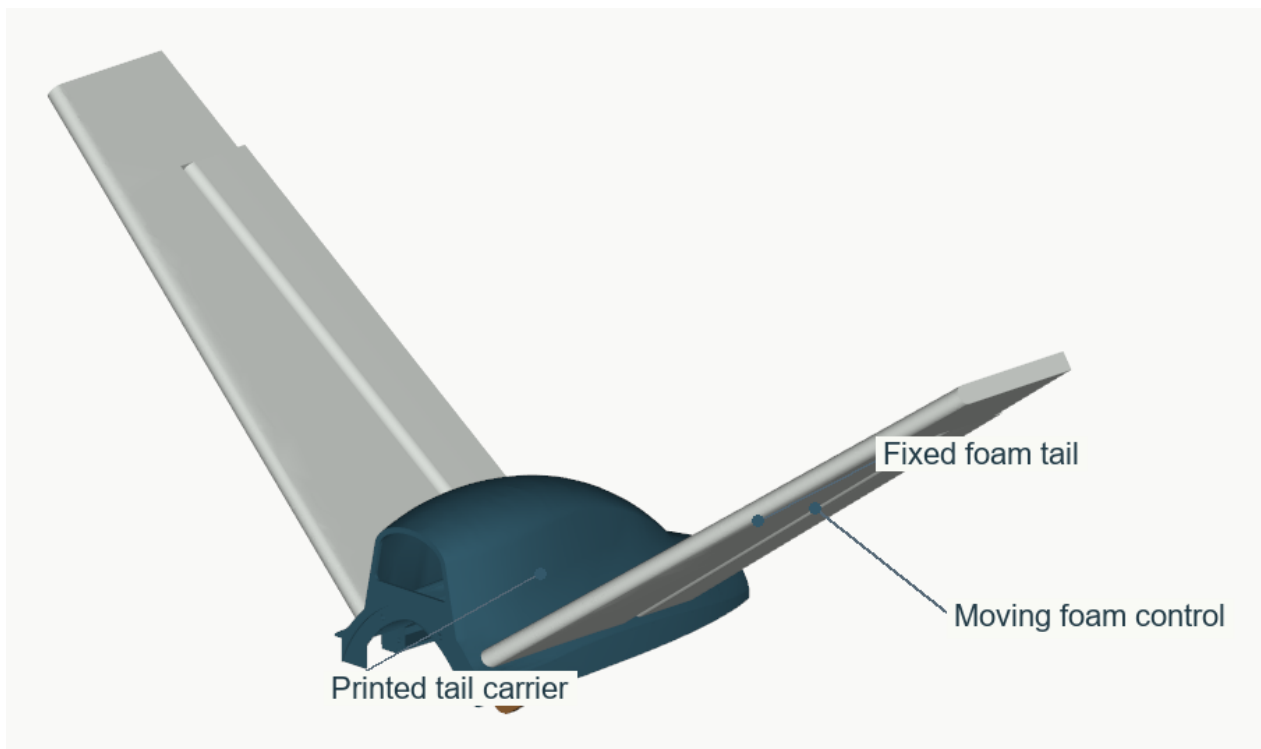
MARK_DO_NOT_CUT lines are hand references. Do not laser through a fold line. Relieve the inner core only as needed while keeping the outer paper continuous. Dry-fit and trim perimeter before bonding.

FW11: tail underside. Trial in paper first: this compound curve has up to 1.315% local flattening strain. If it buckles, use short overlapping darts from the perimeter while keeping one connected piece. Tape the outer seam; do not force a buckle flat.

Moving tails: FW12 right; FW13 left. Laminate two layers per side, then remove the 5 mm trim allowance to the marked outline and carve the rounded leading edge. The marks are shaping references, not extra cuts.

13 foam patterns / 15 physical 5 mm blanks / 3 full 1000 x 600 mm sheets / 5 cutter-bed loads. FW12 and FW13 are cut twice, then laminated into the two 10 mm moving tail controls.

Close the body and form the tail



26 Fit body wraps

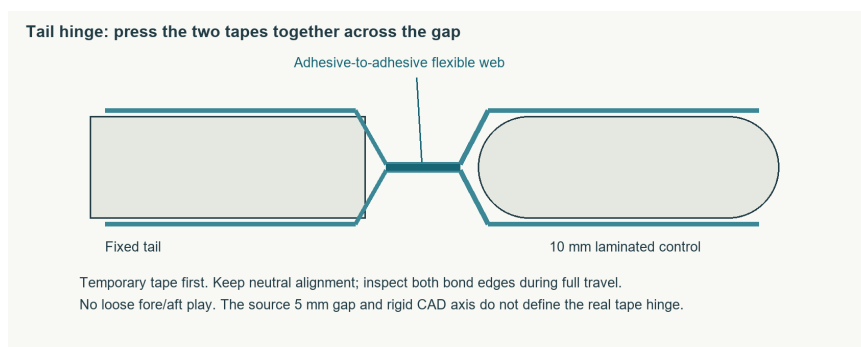
Trial-fit lower, middle and upper body wraps to the frame and printed shells. Use internal seam tape/straps where specified. Leave battery and electronics access serviceable; leave gear openings until the actual gear is positioned.

27 Form the fixed V-tail

Build both fixed tail halves from the R5 patterns and compare them against the printed A12 tail carrier. Keep left/right angles, root seating and trailing edges symmetric. Do not force a warped surface into its socket.

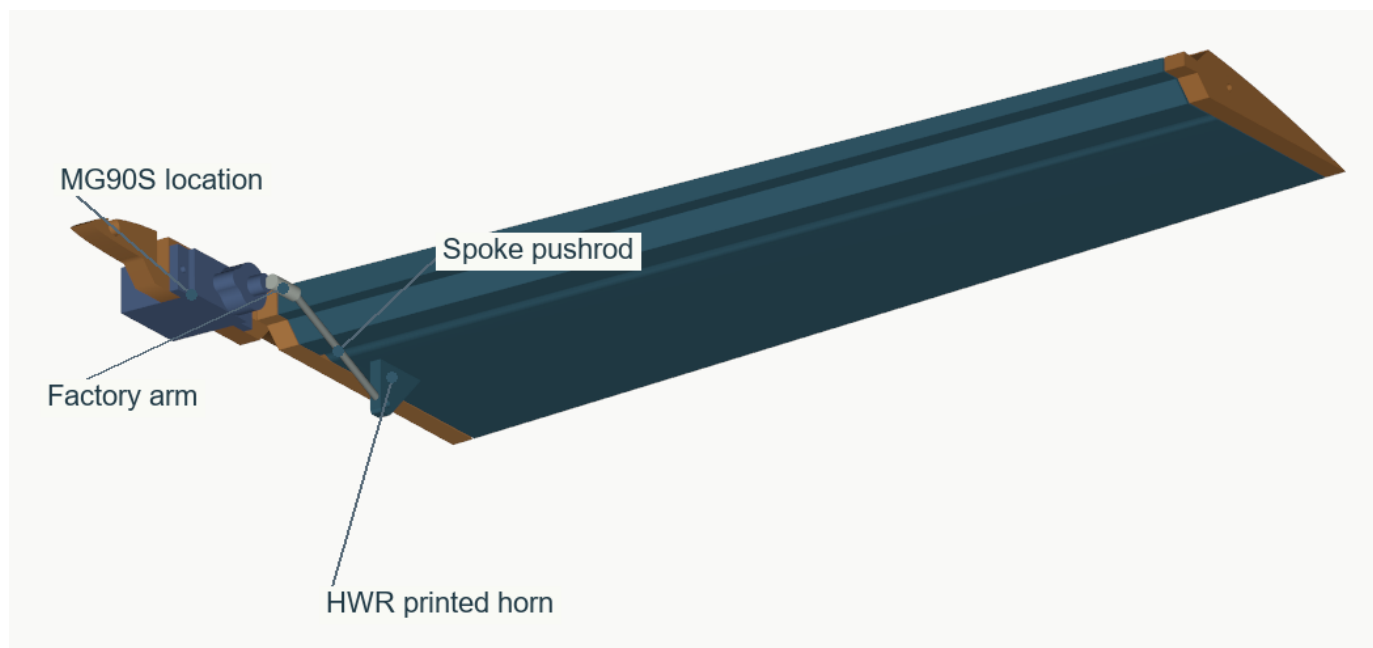
28 Laminate the moving controls

A41 and A60 retain the original 10 mm section. Laminate two 5 mm blanks for each, cure flat, and finish the outline/edge shape against the pattern. Trial the reinforced-tape web hinge shown below at neutral alignment. It must move freely both ways without pulling the foam apart.



Apply reinforced tape on both faces, then press adhesive to adhesive across the gap to make one flexible web. Trial +/-15 degrees without peel, binding or fore/aft slop; this is a bench fit check, not a flight-qualified throw or rigid-axis guarantee.

Set the controls before covering



29 Centre the four servos

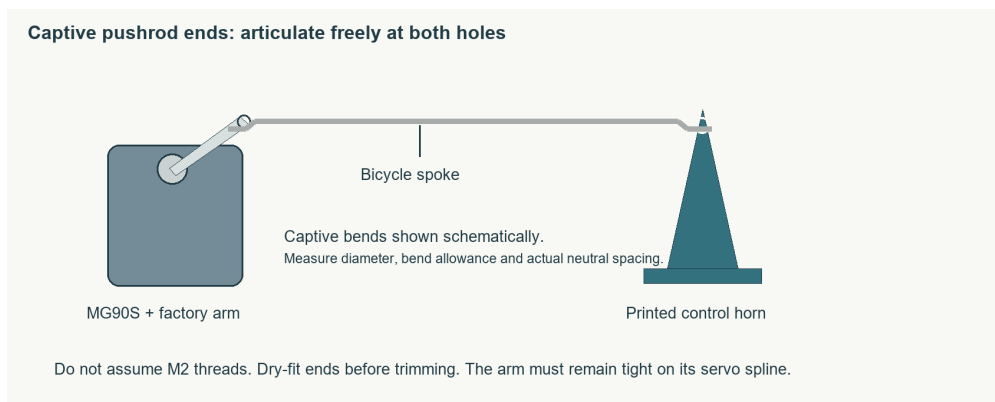
Use the factory servo arms on the MG90S units. Centre them electrically with the propeller removed, then fit the arms. Use S3/S4 for the wing servos and S5/S6 for the tail; verify the final left/right assignment in setup.

30 Fit four horns and four pushrods

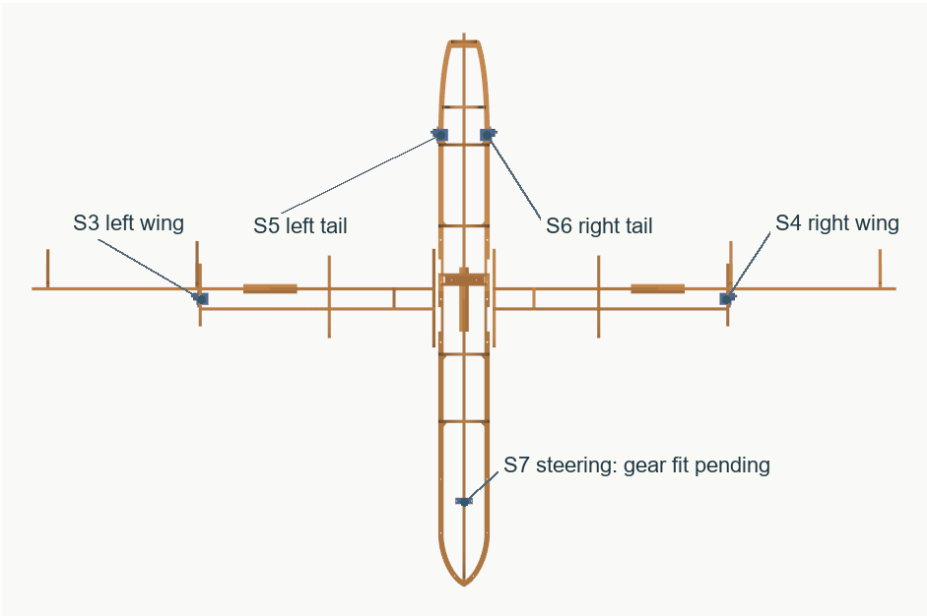
Fit the four printed horns at the intended control points. Use the four bicycle spokes as pushrods, cutting only after both ends and full travel are established. Measure the spoke; do not assume its thread is M2. Use captive ends that articulate without binding.

31 Check every endpoint

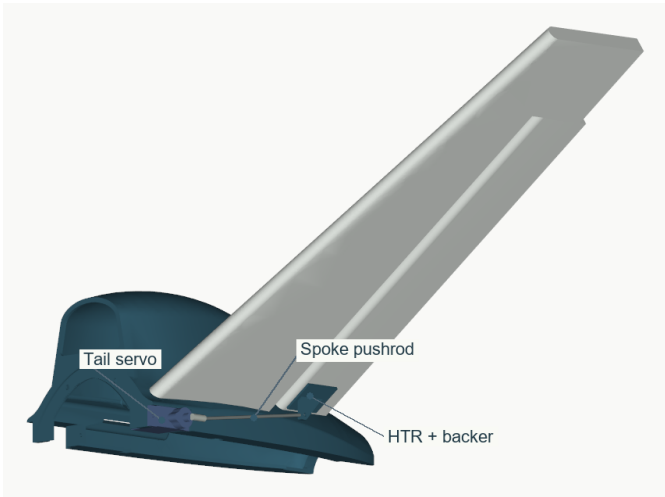
Hold the airframe and apply light control load while moving to both limits. Look for foam flex, horn peel, rubbing, servo buzz and an arm going over centre. Fix the cause mechanically before reducing travel in software. Keep adjustment points accessible.



Servo locations and tail linkage

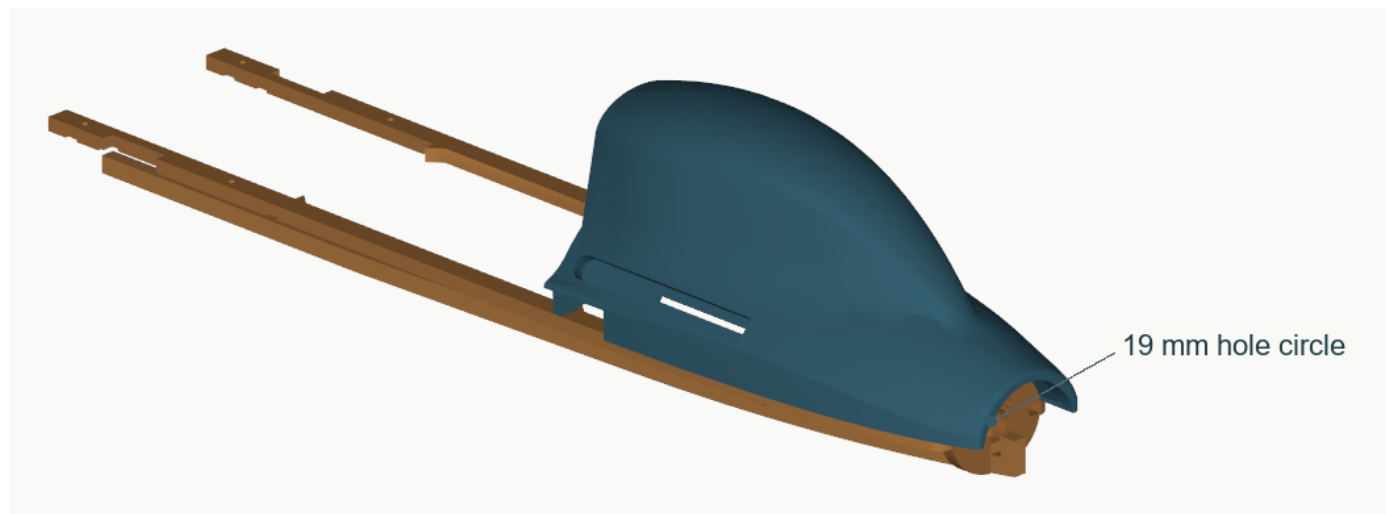


Output / location	Factory-arm shaft target X / Y / Z mm
S3 left wing; S4 right wing	-509 / 106.258 / 12.326; +509 / 106.258 / 12.326
S5 left tail; S6 right tail	-63.85 / 411.10 / 6.215; +63.85 / 411.10 / 6.215
S7 nose steering	Near the original nose-gear station; final position follows the purchased gear and free steering linkage.



Source servo envelopes and actual R5 horn positions shown. Arms use a provisional 10 mm lever radius and straight 2 mm spoke envelope for illustration only. Measure your factory arm hole, spoke and captive bends; do not cut to a length inferred from the picture. Fit the broad tail horn and 1.5 mm backer against the actual foam before bonding.

Fit the motor and owned nose



32 Check the real motor against A26

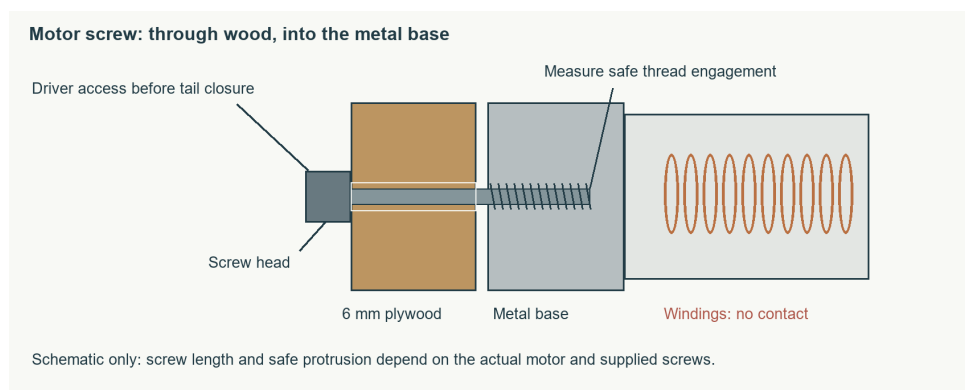
The mount has four 3.2 mm holes on a 19 mm pitch circle, rotated 45 degrees. Place the owned Emax motor against the pattern and confirm the centre opening, bell clearance, wire exit and straight screwdriver path.

33 Use the supplied motor screws carefully

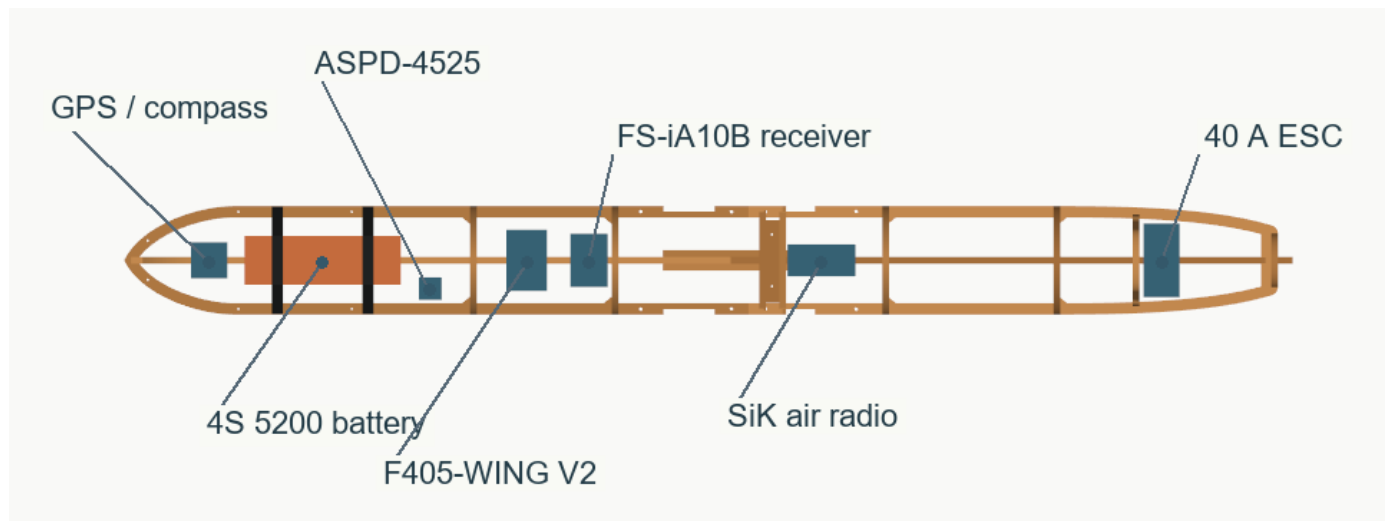
Screws pass through the wood clearance holes and thread into the metal motor base. Measure 6 mm wood thickness, screw length and safe thread engagement; the supplied length is not yet verified. A screw must clamp without bottoming out or reaching windings. Check driver access before tail closure, with the propeller off.

34 Make only the local pitot passage

Reuse N1 unchanged. Mark the pitot and tubing route on the real nose after checking inner clearance and the battery/access path. Drill the smallest practical guided passage and smooth its edge. Support the tube with simple tape/ties; keep the sensing openings clear.



Gear, battery and equipment placement



Placement targets shown as nominal device boxes on the actual frame. Check real plugs, leads, cooling, pad contact and removal access before taping or tying anything in place.

35 Leave gear attachment open until selected

Use the purchased tricycle set and its supplied clamps, collars and fasteners once its dimensions are known. Attachment must spread landing loads into the wood frame, allow steering, and keep the 9-inch prop clear of the ground. The exact attachment method remains owner-selected; foam must not carry the landing load.

36 Restrain the battery positively

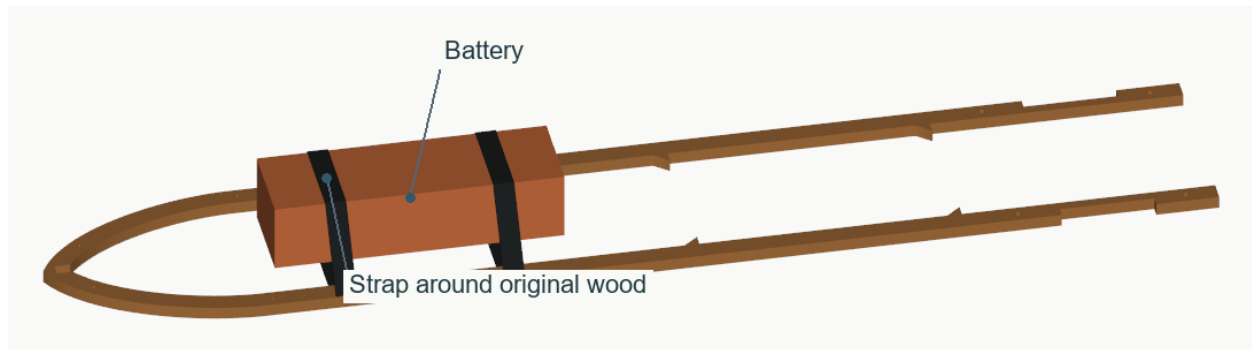
Use a nonslip pad and two independent straps/ties around a sound frame path. Adhesive can locate the pad but cannot be the only battery restraint. Adjust the 512 g battery fore/aft only after weighing and balancing the complete aircraft.

37 Attach electronics simply and serviceably

Use suitable tape, small glue pads and zip ties; no custom equipment mounts are required. Mount the flight controller with its orientation known and protect it from frame vibration. Keep GPS/compass away from motor, power leads and magnets. Separate antennas and route wiring clear of moving controls.

Placement targets and service access

Device; envelope X/Y/Z mm	Target centre X/Y/Z mm	Attachment / access
4S 5200 battery; 43 / 138 / 33	0 / -260 / 18.5	Two Velcro straps around A13 wood; lift nose for access; final CG sets position.
F405-WING V2; 54 / 36 / 13	0 / -80 / 10	Known arrow/orientation; insulated vibration pad; leave USB and plugs reachable.
FS-iA10B; 47 / 33.1 / 14.7	0 / -25 / 10	Keep servo-iBUS connector reachable; separate antennas from carbon and power leads.
M9N-5883; 32 / 32 / 10	0 / -360 / 50	Antenna face upward inside nose; tape/ties; verify compass interference after wiring.
ASPD-4525; 20 / 20 / 12	25 / -165 / -12	Near pitot tube route; avoid kinks; leave both pressure ports accessible.
SiK radio; 28 / 60 / 18	0 / 180 / 14	Serviceable tape/ties; keep antenna clear of carbon and high-current wiring.
40 A ESC; 65 / 32 / 16	0 / 480 / 15	Tape/ties to original frame; expose cooling surface; keep motor wires short and secured.



Strap routing illustration: wrap around the existing wood and battery, not a floating pad. Check load-bearing contact, edge rounding and pack removal on the real frame. The nominal battery bottom is Z2 mm; the shown centre Y-260 mm is provisional until measured CG.

Five servo locations: S3/S4 at the left/right original outer-rib pockets; S5/S6 beside the original tail roots, with arms and horns accessible. S7 near the nose steering linkage only after the bought gear is fitted. Source servo shapes are placement references; measure the actual MG90S body, tabs and arm clearance.

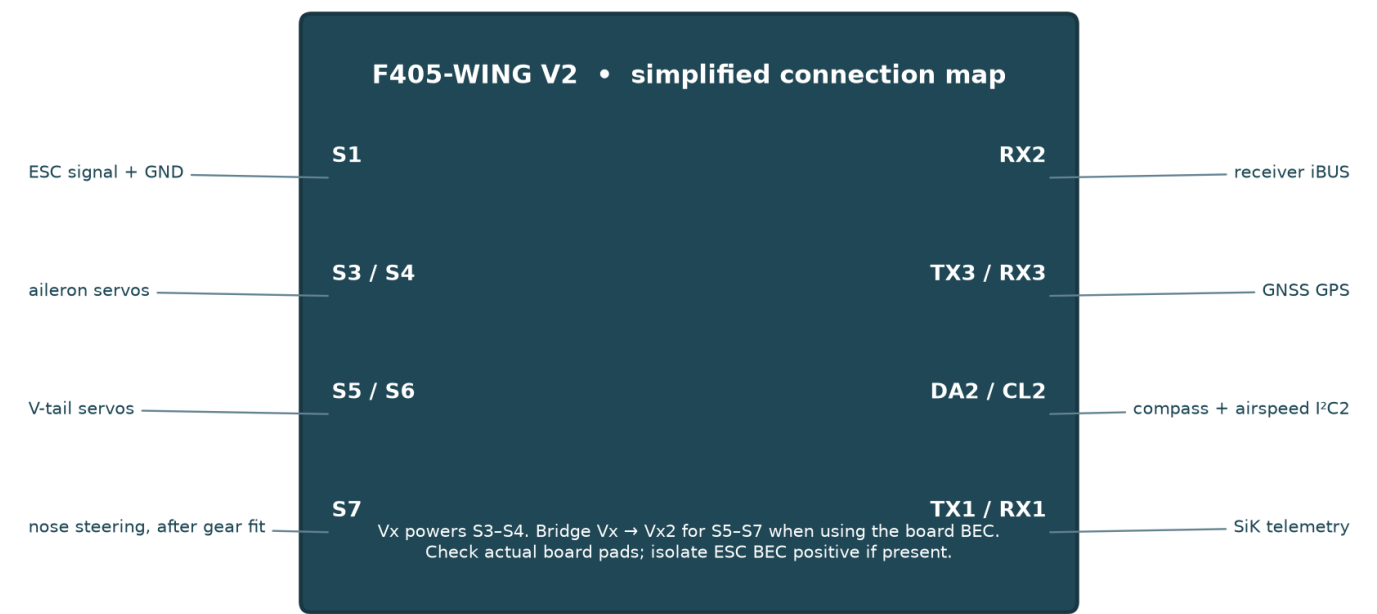
Wire the F405-WING V2

Device	Board connection	Check before power
4S battery / power input	Battery leads to rated board power input through matching connector	Check polarity, wire rating and continuity; insulate joints.
40 A ESC power + motor	ESC thick power pair to rated battery distribution; three motor phase leads to motor	Follow actual V2 pad marks and ESC lead labels. Secure three motor wires.
40 A ESC control	Signal S1 + GND	Isolate any live ESC BEC positive; do not parallel servo supplies.
MG90S wing servos	S3 and S4 / Vx / GND	Vx set to 5 V; confirm polarity on every plug.
MG90S tail servos	S5 and S6 / Vx2 / GND	Bridge Vx to Vx2 for the onboard BEC supply.
Optional steering MG90S	S7 / Vx2 / GND	Only after bought gear fit and steering linkage are proven.
FS-iA10B receiver	Servo iBUS to RX2; rated 5 V + GND	Use servo-iBUS output, not SENS; check polarity.
M9N-5883 GNSS	GPS TX to RX3; GPS RX to TX3; supply + GND	Check actual module supply voltage and orientation.
M9N compass + ASPD-4525	DA2 / CL2, supply and GND	Shared I2C2; no swapped SDA/SCL; avoid long loose leads.
SiK telemetry	Radio TX to RX1; RX to TX1; supply + GND	Check exact radio supply and logic requirements.

Disconnect the propeller for wiring, calibration and first motor-direction tests. Measure the servo rail at 5 V before plugging in the servos. Test simultaneous servo movement under representative load; five servos do not automatically fit the BEC current budget.

Primary source: [Matek F405-WING-V2 product page and pin mapping](#). Onboard Vx: 5 A continuous / use 6 A peak limit from the product specifications. Vx2 is unpowered unless bridged or separately supplied.

Find the pads on the actual board



This connection drawing is illustrative. Use the printed pad labels on your physical V2 board as the final check. The manufacturer layout is [linked below](#); check board revision before soldering.

Find first	Then connect
S1 and G; Vx and Vx2	Throttle signal/ground; 5 V servo power and required Vx-to-Vx2 bridge.
RX2; TX3/RX3; DA2/CL2	iBUS receiver; GPS; shared compass and airspeed I2C2.
TX1/RX1; 5V/G	Telemetry UART and verified supply.

Pin mapping checked against the [Matek F405-WING-V2 product page](#) and [actual layout](#). This schematic is an original explanatory drawing, not a photo of an assembled aircraft.

Configure and calibrate

38 Identify the board and save a baseline

Install the correct firmware target for the actual F405-WING V2. Save parameters before changes. Confirm the displayed board orientation matches nose/roll/pitch movements. Configure the receiver as iBUS on RX2 and inspect every channel.

39 Assign the outputs explicitly

S1 is the ESC throttle output, not the standard Plane aileron default. Assign S3/S4 to the two ailerons and S5/S6 to the two V-tail controls according to their physical sides. Use the flight controller V-tail mix; turn transmitter V-tail mixing off. Optional S7 is GroundSteering (function 26) only after the gear is fitted.

40 Calibrate with the propeller removed

Calibrate accelerometers and compass; set the true mounted compass orientation. Check GNSS lock and telemetry. Calibrate airspeed zero in still air and inspect the tube routing/ports. Set radio endpoints and failsafe; confirm the receiver loses signal as expected.

41 Check power measurements

Select the V2 battery/current monitor settings, then compare voltage and current readings against measurement. Matek lists BATT_AMP_PERVLT = 66.7 for V2; do not copy the V1 scale. Confirm actual servo-rail voltage and supply behavior under simultaneous movement.

ArduPlane parameter names and output numbers are not a substitute for physical direction checks. Read current board and firmware instructions, then verify the actual saved configuration and response on this aircraft.

Sources: Matek F405-WING-V2 mapping; [ArduPilot Plane servo functions](#); [ground steering](#). Dedicated steering also needs GROUND_STEER_ALT > 0 and a working compass.

Prove the controls on the bench

Test	Expected observation
Manual roll right	Right aileron rises and left aileron lowers. Reverse a wrong servo/output, then recheck all modes.
Manual pitch up	Both V-tail trailing edges move toward the pitch-up combination. Verify the mix mechanically and against the firmware tail convention.
Manual yaw	The two tail controls produce the requested yaw combination. Check both yaw directions and combined pitch/yaw endpoints.
Stabilized attitude response	Tilt/roll the aircraft by hand. Surfaces must oppose the disturbance. Check the controller orientation and response sign.
Radio loss / failsafe	With prop removed, turn transmitter off in a controlled test. Verify the configured safe throttle/mode response and recovery.
Full combined control travel	No binding, pin escape, horn peel, rubbing or servo stalling at any combined command.
Harness and connector check	Pull gently on each restraint; flex wings/controls. No wire snags, loose plugs or unsupported solder joints.

Repeat the direction checks after any output reassignment, reverse change, firmware update or linkage change. Use an experienced pilot to verify the V-tail mix and stabilized responses before the first flight.

Record the actual control throws, centre positions, endpoints and firmware settings after the aircraft passes. The guide does not invent a flight-qualified throw or CG from appearance.

Ground tests and balance

42 Weigh and balance complete

Include the flight battery, straps, gear, propeller, wiring and all fasteners. Record total mass and individual left/right balance. Set CG from the final R5 analysis, then physically measure it with the battery restrained in that position.

43 Test the structure progressively

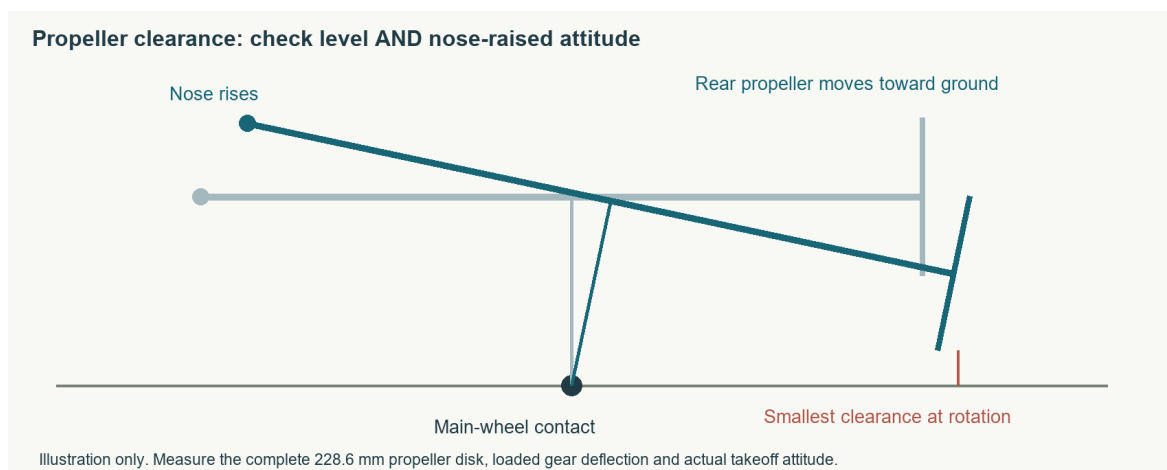
Inspect cured wood laps, wing roots, whole-rod retention, motor plate, tail attachment and gear load paths. Apply representative loads with the airframe supported safely. Stop for cracking, slip, permanent bend or adhesive peel. A CAD intersection test is not a load test.

44 Qualify the exact propulsion set

With safe restraint, clear the propeller plane and measure current, voltage sag, thrust and temperatures using the owned 4S/2807/9045/40 A combination. Stop on excessive current, heat, vibration or loosening. Confirm pusher rotation and propeller orientation before a powered run.

45 Check gear and taxi behavior

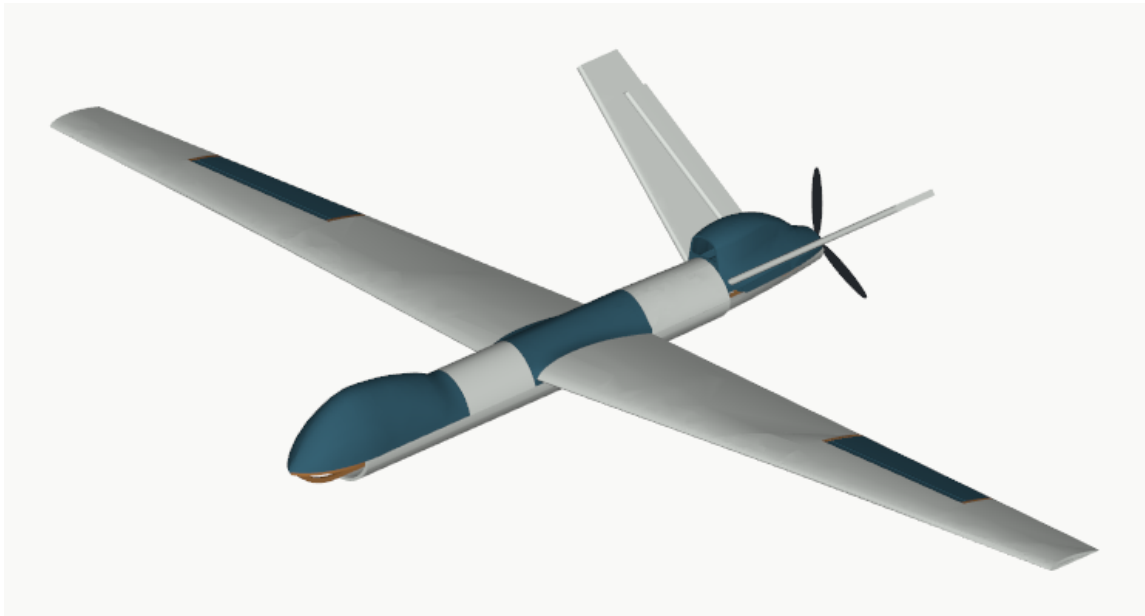
After the selected gear is attached, load the aircraft to flying mass. Verify prop clearance with gear deflection, steering throw, wheel tracking and ground handling. Reinspect the mounts after the test. Select a suitable runway with an experienced pilot.



Tilt the aircraft about the main-wheel contact to its proposed takeoff attitude. Measure clearance to the whole 228.6 mm propeller disk with loaded gear deflection, not just one bottom blade or the level stance. A rear pusher can strike as the nose rises.

No hand launch, takeoff distance, endurance or flight load rating is established by this guide. Record measured results and obtain pilot acceptance before committing to flight.

Before every flight



Check	Confirm
Airframe	No loose laps, cracks, distorted prints, foam separation or softened adhesive.
Wings and controls	Rods and roots retained; pins captive; horns and pushrods secure; full free travel.
Battery and CG	Correct pack secured twice; measured CG accepted; wiring cannot reach propeller or controls.
Gear and propeller	Attachment tight; steering free; prop and hub sound; ground clearance proven.
Electronics	Correct model and modes; voltage/current sensible; GPS/compass/airspeed healthy.
Direction and failsafe	Manual and stabilized responses correct; failsafe tested after changes; range check completed.
Site and pilot	Suitable runway, weather and clear area; experienced pilot agrees aircraft is ready.

After the first ground run and every hard landing, reopen access and inspect high-load joints. Log changes, measurements and failures so the next person can reproduce the accepted setup.

Files and source references

File / group	Purpose
Plywood/REAPER_R5_PLY_PANEL_01–02.dxf	Two laser jobs, 609.6 x 457.2 mm; 37 aircraft pieces plus 2 test coupons; grain along long edge.
Foam/REAPER_R5_FOAM_PANEL_01–05.dxf	Five laser jobs from three full foam sheets; 13 patterns / 15 blanks. Cut layer only.
Print/Individual/*.stl; Print_Batches/*.stl	Original-style printed parts and horns; batch jobs are optional. Reuse the already printed N1.
CAD/Reaper_R5_Assembly.step; Data/Parts.csv	Original-style part placement and exact IDs. Gear is still owner-selected.
Data/Physical_Checks.csv; Data/Calculation_Summary.json	Record real stock, mass, balance, propulsion, joints and control proof.

Construction source

[Michael Rehtin: Building a DIY REAPER Drone... Ended Badly](#). Original supplied assembled/disassembled CAD is the geometry reference. R5 retains that construction approach with local changes for owned stock, carbon rods, hinge fit and motor pattern.

Electronics primary references

Matek F405-WING-V2 • M9N-5883 • ASPD-4525 • FlySky FS-iA10B • ArduPilot Matek board notes. Research checked 24 September 2026; verify firmware and received hardware revisions.

Remaining physical decisions: exact printer identity, purchased gear and attachment, actual pin/rod/stock fits, measured motor screw engagement, cured joint proof, finished mass/CG, propulsion results and pilot acceptance. Keep those results with the aircraft.